Academic Curriculum Vitae
Dirk Gütlin

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for more information and work examples, visit my blog: https://digyt.github.io



Studies

Freie Universität Berlin, Center for Cognitive Neuroscience Berlin
MAR 2023 - CURRENT

Research Associate/PreDoc, Prediction & Memory Group:
Biologically plausible computational models of neural learning and inference.
Supervisors: Prof. Ryszard Auksztulewicz, Prof. Felix Blankenburg.

University of Osnabrück
OCT 2019 - JUN 2022

Cognitive Science (Master of Science) - with honors
Thesis title: 'Brain Reading using Transformers'
Supervisors: Prof. Peter König, Prof. Tim Kietzmann

University of Salzburg
OCT 2016 - JUN 2019

Psychology (Bachelor of Science)
Thesis title: 'The Salzburg Visual Field Trainer: A Virtual Reality Application for Visual Restoration Therapy'
Supervisors: Stefan Hawelka, Michael Leitner

University of Koblenz-Landau
OCT 2015 - JUN 2016

Mathematics, Psychology (Bachelor of Science)

Other Experience / Associations

Bernstein Center for Computational Neuroscience Berlin
Oct 2024 -

Associate member, BCCN doctoral program.
Supervisors: Prof. Felix Blankenburg.

University of Osnabrück, Institute of Cognitive Science
MAR 2022 - Sept 2022

Research Assistant, Machine Learning Group:
Neural Decoding using Transformer Neural Networks; Streamlining Dataset accessibility for Machine Learning use.
Supervisors: Prof. Tim Kietzmann.

Interests

Computational Neuroscience
Machine & Deep Learning
Time Series & Statistics
Complex Systems
Data Analysis

Programming Languages

Python
MATLAB
R
C#
JS/HTML/CSS

University of Osnabrück, Institute of Cognitive Science
MAR 2020 - MAR 2022

Research Assistant, Neurobiopsychology:
Feature classification for eye tracking data; Concussion prediction based on eye tracking; Representational Similarity Analysis with EEG data; Investigation of automated noise removal algorithms for EEG data.
Supervisors: Artur Czeszumski, Prof. Peter König.

University of Osnabrück, Institute of Cognitive Science
MAR 2020 - SEPT 2021

Research Assistant, Cognitive Modelling Group:
Interactive online experiments for investigating language pragmatics;
Teaching Assistant: 'Experimental Psychology Lab' - teaching experimental design, implementation, and statistical evaluation.
Supervisor: Prof. Michael Franke.

Google Summer of Code Project with Stipend for Python Software Foundation/ MNE-Python
MAY 2019 - AUG 2019

Google Summer of Code Participation, submitting the project:
'MNE-Python: Improve Time Frequency Analysis in Source Space' for M/EEG data analysis software MNE-Python.
Supervisors: Alexandre Gramfort, Joan Massich.

Centre for Cognitive Neuroscience Salzburg
SEPT 2018 - AUG 2019

Research Assistant, Neurocognition Lab:
Development of research applications, Eye-Tracking measurements.
Supervisors: Michael Leitner, Prof. Florian Hutzler.

University of Salzburg, Department of Psychology
MAR 2018 – AUG 2018

Tutor: 'MATLAB - Advanced Applications':
Neurophysiological data processing with FieldTrip, experimental Designs with Psychtoolbox.
Supervisor: Gianpaolo Demarchi.

Christian-Doppler-Klinik Salzburg
DEC 2017 – SEPT 2018

Intern, Department of Neurology:
Epilepsy monitoring, EEG measurement, automated epileptic spike detection.
Supervisors: Yvonne Höller, Aljoscha Thomschewski.

University of Salzburg, Department of Psychology
NOV 2017 – JUL 2018

Intern, Salzburg Brain Dynamics Lab:
MEG measurement support, MEG Data Analysis (Time-Frequency Analysis, MVPA, Speech Reconstruction using ANN).

Supervisor: Prof. Nathan Weisz.

Selected Publications

Gütlin, D., Auksztulewicz, R., & Doerig, A. (2026). Local learning is all you need: Brain-aligned predictive coding networks [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2610.37327

Gütlin, D., Kittelmann, D., & Auksztulewicz, R. (2026). Predictive coding networks capture human neural representations missing in supervised DNNs. *bioRxiv*, 2026-09.

Gütlin, D., & Auksztulewicz, R. (2025). Predictive Coding algorithms induce brain-like responses in Artificial Neural Networks. PLOS Complex Systems, 2(11), e0000076.

Gütlin, D. C.*, McDermott H. H.*, Grundei M., & Auksztulewicz, R. (2023). Model-based approaches to investigating mismatch responses in schizophrenia. Manuscript submitted for publication.

Leitner, M. C., Gütlin, D. C., & Hawelka, S. (2021). Salzburg Visual Field Trainer (SVFT): A virtual reality device for (the evaluation of) neuropsychological rehabilitation. PLOS ONE. DOI: 10.1371/journal.pone.0249762

Published Software

CatEyes - Unified Categorization for Eye Tracking

Python Toolbox for simple and uniform application and visualization of popular feature classification algorithms for eye tracking.
Repository: https://github.com/DiGyt/cateyes

ASRpy

Python implementation of the Artifact Subspace Reconstruction Cleaning algorithm for M/EEG cleaning.
Repository: https://github.com/DiGyt/asrpy

pymatreader

Python Toolbox for uniform translation of MATLAB (.mat) files into Python.
With Thomas Hartmann.
Repository: https://gitlab.com/obob/pymatreader

Awards and Scholarships

AI for Climate Hackathon - Category Winner: Cryosphere

JAN 2021

Winning Project for “AI for Climate” Hackathon 2021 hosted by Potsdam Climate Institute and University of Osnabrück. Predicting global snow sheet development based on climate data.

Google Summer of Code Stipend

MAY 2019 - AUG 2019

Stipend issued by Google LLC for contribution to Open Source Software.